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On the exactness of categories of internal groups

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The category $\mathrm{Hopf}_k^{\mathrm{coc}}$ of cocommutative Hopf algebras over a field k forms a semi-abelian, in particular Barr-exact, category [2]. It is the category $\mathsf{Grp}(\mathcal{C})$ of internal groups in $\mathcal{C} = \mathsf{Comon}_k^{\mathrm{coc}}$, which is itself *not* exact. The argument for exactness requires a classical result by Newman [3].

Let $\mathcal{C}$ be a finitely complete category with reflexive coequalizers that are preserved by the binary product componentwise. For an internal group h, let $\mathcal{C}^h$ be the category of h-actions. There is a surjective Galois connection $\mathrm{Sub}_{\mathsf{Grp}}(h) \rightleftarrows \mathrm{Quot}_{\mathcal{C}^h}(h)$ from subgroups $g \leq h$ to regular quotients of h in $\mathcal{C}^h$. A subgroup g is sent to its coset quotient $h \to h/g$, the coequalizer of the *coset (equivalence) relation* $h \times g \rightrightarrows h$, and a regular quotient $q\colon h \to m$ to the stabilizer $q^{-1}(q(e))$ of its base point.

A subgroup $g \leq h$ is a stabilizer precisely when its coset relation is effective. When all coset relations are effective, $\mathsf{Grp}(\mathcal{C})$ is exact if and only if it is regular, which is equivalent to a condition on composites of regular epimorphisms. In this way, Newman's correspondence becomes a categorical criterion for the exactness of $\mathsf{Grp}(\mathcal{C})$. We also show that regular monomorphisms of $\mathsf{Grp}(\mathcal{C})$ are stabilizers. If $\mathcal{C}$ is cartesian closed and every object embeds into an internal group, then conversely every stabilizer is regular. As a consequence, every monomorphism of cocommutative Hopf algebras is regular.

Affine group schemes also form an exact category [4, 1], but the technique does not apply as the coset relations need not be effective (the Borel subgroup of SL_2).

Keywords: Internal groups; Barr-exact categories; semi-abelian categories; cocommutative Hopf algebras; affine group schemes
Subject area: Theory of Categorical Structures
MSC2020: 18E13, 16T05, 14L15

References

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[3] K. Newman, A correspondence between bi-ideals and sub-Hopf algebras in cocommutative Hopf algebras, *J. Algebra* **36** (1975).

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